Anna Raymaker

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Research Overview

I am an interdisciplinary researcher who uses empirical methods to study the security and reliability of critical infrastructure. I combine large-scale measurement, data analysis, machine learning/AI, and human-centered methods to understand real-world threats and improve the security of transportation, energy, and other safety-critical systems.

Education

Ph.D. in Electrical and Computer Engineering — Fall 2022 - Present
Expected graduation — **May 2028**
Georgia Institute of Technology — Atlanta, GA
Cyber-Physical Systems Security Lab
Advisor: Professor Saman Zonouz

M.S. in Electrical and Computer Engineering — Spring 2025
Georgia Institute of Technology — Atlanta, GA

B. S. in Computer Engineering — Spring 2022
University of Florida — Gainesville, Florida

Honors, Awards, & Funding

ACM CCS Student Travel Grant — Fall 2026
Awarded $1,000 to support travel to the ACM Conference on Computer and Communications Security (CCS 2026).

2026 Student Leader Award [link] — Spring 2026
Awarded by Georgia Tech to students demonstrating outstanding leadership in advancing student engagement.

Spark Award Scholarship [link] — Fall 2025
Awarded to Georgia Tech graduate students demonstrating outstanding leadership in advancing student engagement in energy research and broader service impact.

President's Fellowship — 2022-2026
Offered to the top 10% of the Georgia Institute of Technology's doctoral program applicants.

Commencement Speaker at the University of Florida [link] — Spring 2022
(52:00 - 57:00 approx.)

Presidential Service Award at the University of Florida — Spring 2022

National Merit Scholar — 2018

Benacquisto, Bright Futures, and Vicar's Landing Scholarships — 2018 - 2022

Publications

Raymaker, A., Von Brock, R., Pickren, R., Chhotaray, A., Li, F., Zonouz, S., Beyah, R., "She Spoofed Sea Ships by the Sea Shore: Measuring Large-Scale GPS Spoofing in Global Maritime Traffic," to appear in *IEEE Symposium on Security and Privacy (IEEE S&P)*, 2027

Raymaker, A., Talkington, S., Ryu, Z., Asiamah, R., Abukhousa, E., Ashebo, B., Chhotaray, A., Molzahn, D., Li, F., Zonouz, S., Beyah, R., "Grid Trouble in Paradise: Uncovering Vulnerable Distributed Energy Resources and Their Grid-Level Risks," to appear in *The ACM Conference on Computer and Communications Security (CCS)*, 2026

Raymaker, A., Meller, A., Duan, H., Okhravi, H., Zonouz, S., Kuhn, S., "A Sea of Lies: GPS Spoofing and the Emerging Maritime Cyber Battlefield," *IEEE Security & Privacy*, 2026.

Raymaker, A., Kumar, A., Wong, M., Pickren, R., Chhotaray, A., Li, F., Zonouz, S., Beyah, R., "A Sea of Cyber Threats: Maritime Cybersecurity from the Perspective of Mariners," in *The ACM Conference on Computer and Communications Security (CCS)*, 2025

Abukhousa, E., **Raymaker, A.**, Chhotaray, A., Molzahn, D., Zonouz, S., Meliopoulos, S., "From Caveman to COBA: Coordinated Optimization-Based Attacks on Grid-Edge Smart Inverters in DER-Rich Power Systems," to appear in *IEEE PES Grid Edge Technologies Conference & Exposition*, 2027.

Von Brock, R., **Raymaker, A.**, Chhotaray, A., Li, F., Zonouz, S., Beyah, R., "Batten the Hatches: Cybersecurity with Military Mariners," to appear in *The ACM Conference on Computer and Communications Security (CCS)*, 2026

Kumar, A., **Raymaker, A.**, Specter, M., "Security and Privacy Analysis of Tile's Location Tracking Protocol," *USENIX Security Symposium*, 2026

Ashebo, B., **Raymaker, A.**, Talkington, S., Asiamah, R., Chhotaray, A., Zonouz, S., Molzahn, D., "Analyzing the Cyber Resilience of Distribution Systems with Vulnerable Inverter-Based Resources," to appear in *Hawaii International Conference on System Sciences (HICSS)*, 2027

*Pickren, R., ***Raymaker, A.**, Chhotaray, A., Li, F., Zonouz, S., Beyah, R., "Maritime Vessel Networks as an Attack Surface: Architecture, Exploitation, and Global Impact," *in submission*, 2026

Raymaker, A., Meller, A., Duan, H., Zonouz, S., Okhravi, H., "This Is How [Ghost Ships] Disappear: Security Analysis of the Automatic Identification System," *in submission*, 2026

Ryu, Z., Chung, S., Karim, M., **Raymaker, A.**, Jodha, K., Chaturvedi, Y., Mertoguno, S., "Beware EviLLM: Enabling Vulnerability Injection via Large Language Models," *in submission*, 2026

*Authors contributed equally.

Relevant Coursework

Graduate Machine Learning, Advanced Computer Security, Empirical Computer Security, Cybersecurity of Drones, Network Security, Advanced Malware Analysis, Linear Systems and Controls, Advanced Programming Techniques

Technical Skills

Python (NumPy, pandas, scikit-learn, scipy), C, C++, supervised classification, feature engineering, NLP/TF-IDF, LLM prompting and zero-shot classification, statistical modeling, anomaly detection, Internet-scale data pipelines, distributed systems analysis, version control (Git)

Research Experience

Graduate Research Assistant - CPSec Lab Spring 2024 - Current
Georgia Institute of Technology Atlanta, GA

Energy and Critical Infrastructure Security (Department of Energy Funded)

- Built a supervised classification pipeline (Random Forest, 24 engineered features, discriminative TF-IDF vocabulary selection) to identify 64K exposed solar energy devices across 1.9M Internet hosts, achieving 98%+ precision via feature engineering, consensus voting, and threshold tuning (CCS 2026).
- Designed an end-to-end data pipeline integrating Internet-scale scanning (Censys), LLM-assisted zero-shot labeling (Llama 3.3:70b), and deterministic fingerprint extraction across 34 vendors to generate training data without manual annotation.
- Led a team of four master's students in the discovery and analysis of exposed DERs.

Maritime and Transportation Cybersecurity

- Built a large-scale maritime data analysis tool to detect GPS spoofing by combining Kalman filtering, geodesic deviation checks, and unsupervised clustering across AIS data from 367,000+ vessels; identified 22 spoofing zones worldwide, 9 previously unreported (IEEE S&P 2027).
- Led a user study identifying key cybersecurity gaps in mariner mental models and training (CCS 2025).

• Built a maritime cybersecurity testbed to evaluate cyber threats and defenses.
• Contributed to a research project on AIS-based malware propagation, assisting in the development and evaluation of situational message-based attack vectors (paper under submission).
• Reverse engineered NMEA 2000 messages over CAN bus to support analysis of vessel controllers.

Binary Analysis for Software Supply Chain Security (Department of Defense Funded)
• Applied machine learning models (e.g., Trex, GNNs) for binary similarity analysis to support automated Bill of Materials (BA-BOM) extraction in large-scale binaries.

Graduate Research Collaborator - D2I Lab Spring 2024
Georgia Institute of Technology Atlanta, GA
• Led a project on data preprocessing pipelines for large-scale ML analytics.

Graduate Research Collaborator - COEUS Fall 2023
Georgia Institute of Technology Atlanta, GA
• Helped develop efficient binary analysis tools, experimenting with regex and code lifters.

Graduate Research Assistant - CyFI Lab Fall 2022 - Fall 2023
Georgia Institute of Technology Atlanta, GA
• Studied web-based malware, focusing on botnet hijacking and remediation.

Undergraduate Research Assistant - ARC lab August 2021 - May 2022
University of Florida Gainesville, Florida
• Developed an FPGA-accelerated genetic programming tool in VHDL.

Undergraduate Research Assistant - FICS lab June 2020 - Feb 2022
University of Florida Gainesville, Florida
• Conducted fault injection attacks on FPGAs to assess FSM vulnerabilities.

Undergraduate Research Assistant - SWAMP lab August 2019 - 2020
University of Florida Gainesville, Florida
• Analyzed SiGe diffusion using TCAD modeling.

Teaching

Head Graduate Teaching Assistant, Georgia Institute of Technology
ECE 3030: Physical Foundations of Computer Engineering Spring 2024

Graduate Teaching Assistant, Georgia Institute of Technology
ECE 6100: Advanced Computer Architecture Fall 2023

Undergraduate Teaching Assistant, University of Florida
EEL 4712: Digital Design Fall 2021 - Spring 2022
EEL 4744: Microprocessor Applications Fall 2021
EEL 3701C: Digital Logic and Computer Systems Spring - Summer 2021

Mentoring

Zeezoo Ryu Summer 2026 - Present
Solar energy device exposure measurement; third author, ACM CCS 2026

Ryan Von Brock Fall 2024 - Present
Maritime user studies; first author of follow-up study, ACM CCS 2026; second author, IEEE S&P 2027

Team of four M.S. students Fall 2025
Led discovery and analysis of Internet-exposed energy devices

Invited Talks

Guest Lecture for Cybersecurity of Drones class (Atlanta, GA) Feb 2026
Presented research on transportation and energy cybersecurity to a class of graduate students at Georgia Tech

SecurityWeek ICS Cybersecurity Conference (Atlanta, GA) Oct 2025
From Ship to Shore: Real-World Threats and Zero-Day Attacks in Maritime Operational Technology
icscybersecurityconference.com

ACM CCS 2025 Conference Presentation (Taipei, Taiwan) Oct 2025
A Sea of Cyber Threats: Maritime Cybersecurity from the Perspective of Mariners

Guest Lecture for Critical Infrastructure Security class (Atlanta, GA) Oct 2025
Presented research on maritime cybersecurity to a class of graduate and undergraduate students at Georgia Tech

Fireside Chat with Dmitri Alperovitch (Atlanta, Georgia) Sept 2025
As President of the GSA for the School of Cybersecurity and Privacy (SCP), I hosted distinguished alumnus Dmitri Alperovitch (co-founder of CrowdStrike) for a fireside chat with Georgia Tech graduate students, discussing his career and insights on cybersecurity.

Georgia Institute of Technology - CPSec Lab Testbed Tour Aug 2025
Demonstrated the maritime cybersecurity testbed and research to faculty members, the School of Cybersecurity and Privacy (SCP) Chair, and distinguished visitors including John Tien (former Deputy Secretary of DHS) and Ann Dunkin (former CIO, U.S. Department of Energy).

TEDxGeorgiaTech Talk Mar 2025
The Invisible War: Protecting Our Future from Cyber Threats
tedxgeorgiatech.com/conference2025

Media Coverage & Writing

Georgia Tech College of Computing October 2026
"When a Ship's Systems Lie, Cybersecurity Gets Complicated"
Featured for research on cybersecurity threats, training gaps, and operational security challenges among military mariners.

MIT Lincoln Laboratory September 2026
"Summer Research Interns Take on National Security Challenges"
Featured for research on security vulnerabilities in the Automatic Identification System (AIS).

Les Echos April 2026
"Quand le GPS perd le nord"
Interviewed as an expert on GPS spoofing and maritime navigation security.

Georgia Tech College of Computing March 2026
"Researchers Find Training Gaps Impacting Maritime Cybersecurity Readiness"
Research coverage of my work on cybersecurity preparedness among commercial mariners.

The Conversation March 2026
"When GPS Lies at Sea: How Electronic Warfare Is Threatening Ships and Their Crews"
Author. Featured as The Conversation's leading article and republished by more than 25 outlets.

WIRED September 2025
"Tile Tracking Tags Can Be Exploited by Tech-Savvy Stalkers, Researchers Say"
Research on security and privacy vulnerabilities in Tile tracking tags featured in WIRED.

Professional Experience

Student Technical Assistant August 2026 - Present
MIT Lincoln Laboratory, Secure Resilient Systems and Technology Group Lexington, MA
• Conducting follow-up research on cybersecurity vulnerabilities and defenses in maritime and energy systems.

Summer Research Intern May 2026 - August 2026
MIT Lincoln Laboratory, Secure Resilient Systems and Technology Group Lexington, MA
• Investigated AIS transponder security in maritime systems, identifying vulnerabilities and evaluating threats to safety-critical vessel communications.
• Research resulted in a paper submitted to NDSS 2027 and an article in IEEE Security &

Privacy Magazine.

FPGA Electrical Engineering Intern May 2021 - August 2021
L3Harris Melbourne, Florida
• Developed SpaceVPX test card in Vivado SPI IP.
• Defined FPGA requirements & collaborated with a multi-specialty engineering team.

Service

Program Committee
USENIX Security Symposium 2027

External Reviewer
IEEE Transactions on Information Forensics and Security (TIFS) 2026
IEEE Security and Privacy (S&P) Magazine 2026
Conference on Computer and Communications Security (CCS) 2023
European Symposium on Research in Computer Security (ESORICS) 2023
IEEE S&P Workshop on Language-Theoretic Security (LangSec) 2023
IEEE International Conference on Trust, Privacy and Security (TPS) 2022

Industry Standards
Member, NMEA Cloud Security Committee May 2026 - Present

Extra-Curricular Activities

President of SCP Graduate Student Association (GSA) May 2025 - May 2026
Georgia Institute of Technology Atlanta, Georgia
• Led the executive board and oversee all GSA initiatives across the SCP.
• Planned and organize inter-college and campus-wide engagement events for the COC.
• Serve as the sole student representative on the SCP Chair Search Committee, ensuring graduate student perspectives are included in the hiring process.

Member of Graduate Women in the College of Computing August 2024 - Present
Georgia Institute of Technology Atlanta, Georgia
• Build community among women graduate students through events and conversations on the experiences of women in computing.

Student and Faculty Affairs Chair for SCP GSA May 2024 - April 2025
Georgia Institute of Technology Atlanta, Georgia
• Organized faculty-student engagement events.
• Advocated for a graduate student ombudsperson and mental health resources.

Presencia Volunteer May 2024 - August 2024
Presencia Atlanta, Georgia
• Assisted immigrant youth in nutrition & fitness programs.

Engineering Ambassador January 2021 - May 2022
University of Florida Gainesville, Florida
• Gave tours for the college and acted as a student engineering leader.

IEEE HKN August 2020 - May 2022
University of Florida Gainesville, Florida
• Held exam reviews and tutored students in the ECE department.

Alpha Phi Omega August 2018 - May 2022
University of Florida Gainesville, Florida
• Volunteered for over 1,000 hours at various local organizations (e.g., the Humane Society, Salvation Army, UF student gardens, CADE Museum for Science and Creativity, local nursing homes).